StudyAI

AI-Based Study Assistant

User & Setup Guide

How to use every feature, step by step, plus how to run, configure and deploy the app.

Live app: https://studyai-production-b064.up.railway.app

Built with Flask, PostgreSQL/SQLite and the Google Gemini API

Contents

Part	What it covers
1. Overview	What StudyAI does and what you need to start
2. Getting started	Create an account, sign in, sign out
3. Using each feature	Dashboard, AI Tutor, Study Notes, Quiz Generator, Flashcards, Study Planner, Settings
4. Limits and small details	Character limits, what is saved automatically, tips
5. Troubleshooting	Common messages and what to do
6. Owner guide: run it locally	Install, configure and start the app on your computer
7. Owner guide: deploy on Railway	Database, environment variables, start command
8. Owner guide: accounts and sign-in history	The admin page, why passwords cannot be shown, browsing the database
9. Owner guide: Gemini API keys and quota	What to do when the free tokens run out
10. Project reference	Files, environment variables, known limitations

1. Overview

StudyAI is a web app for university students. It uses Google's Gemini AI to answer questions, summarise notes, build quizzes and flashcards, and it keeps a personal study planner and progress dashboard. Every person has their own account, and nobody else can see your notes, quizzes, tasks or decks. Accounts are either **Students** (everyone who signs up) or **Admins** (who can also manage accounts).

Feature	What it does
Dashboard	Shows tasks completed, quizzes taken, average quiz score, study streak and a weekly activity chart.
AI Tutor	Chat with Gemini about any study topic. It remembers the conversation, so follow-up questions work.
Study Notes	Paste notes and get a summary, key points, a simple explanation, a mini quiz or flashcards.
Quiz Generator	Create a multiple-choice quiz from a topic, pasted text or an uploaded PDF, then take it and see your score.
Flashcards	Generate flip-style flashcard decks and study them; decks are saved for later.
Study Planner	Add tasks with subject, due date and priority, tick them off, delete them.
Settings	Light/Dark theme, study reminders, password change and recovery code, and a button to erase your saved data.

What you need (as a user)

- A modern web browser (Chrome, Edge, Firefox or Safari).

- An internet connection - the AI features call Google's servers.
- The app's web address. Nothing to install.

2. Getting started

Create an account

1. Open the app's web address. You land on the home page.
2. Click **Get Started** (top right). The **Create Account** page opens.
3. Type a **Username** (3 to 32 characters), a **Password** (at least 6 characters) and the same password again in **Confirm password**, then click **Sign Up**.
4. The next page shows your **recovery code** (for example K7QM-2XPD-9WHF-4BNR). **Save it now** - copy it, download it as a text file, or write it down. It is shown only once. Tick 'I have saved my recovery code', then click **Continue to sign in**.

> **Tip:** If you see 'Username already exists', someone already uses that name. Pick another one. 'Passwords do not match' means the two password boxes differ.

Sign in and sign out

1. On the **Sign In** page enter your username and password and click **Sign In**.
2. You arrive at your **Dashboard**. The top menu now shows Dashboard, AI Tutor, Study Notes, Quiz Generator, Flashcards, Study Planner, Settings and Logout.
3. When you are done, click **Logout**. On a shared computer, always log out.

> **Tip:** Passwords and recovery codes are stored as one-way hashes, so nobody (including the app owner) can read them.

Forgot your password?

1. On the Sign In page click **Forgot password?**.
2. Enter your **username**, your **recovery code** (capitals, lowercase and spaces do not matter) and a new password twice, then click **Reset password**.
3. You see 'Password updated' and a **new** recovery code. The old code stops working, so save the new one.
4. Sign in with your new password.

> **Important:** You get a recovery code when you sign up. If your account is older, or you lost the code, sign in and open **Settings, Account** to create one. Without a saved code (and without being able to sign in) a password cannot be reset - there is no email reset - so keep the code safe.

> **Tip:** After 5 wrong passwords (or wrong recovery codes) for the same username, sign-in is paused for 15 minutes. A successful password reset lifts the pause straight away.

3. Using each feature

3.1 Dashboard

Your home base after signing in. Every number and list on it comes from your own activity, so a brand-new account starts empty and fills up as you use the app:

- **Tasks completed** and **Quizzes completed** - simple counters.
- **Average quiz score** - the mean percentage of all quizzes you finished.
- **Overall task progress** - the share of your planner tasks that are done, with a label: 'On track' (70% or more), 'Keep going', or 'No tasks yet'.
- **Study streak** - consecutive days on which you did something (finished a quiz, completed a task, saved a note or generated a flashcard deck). Missing a day resets it, but a streak from yesterday stays alive until the end of today.
- **Weekly Study Activity** - bars for the last 5 days (quizzes plus tasks you completed that day).
- **Quiz Score Trend** - a line chart of your last 10 quiz scores (appears after your first quiz).
- **Study Progress by Subject** - for each subject in your planner, tasks done out of total.
- **Pick Up Where You Left Off** - shortcuts to your latest flashcard deck, saved note and quiz.
- **Recent Activity** - your most recent quiz results.
- **Quick Actions** - shortcuts to Ask AI Tutor, Create Quiz, Summarize Notes and Create Flashcards.
- **Today's Study Tasks** - planner tasks due today. Click a task to tick it off; the change is saved immediately.
- **Upcoming Deadlines** - your next unfinished tasks, plus up to two overdue ones marked in amber.

3.2 AI Tutor (the chat box)

1. Click **AI Tutor** in the menu.
2. Type your question in the **Your Question** box (for example: 'What is the difference between mitosis and meiosis?'). A counter in the corner shows how many of the 2,000 allowed characters you used.
3. Click **Ask Gemini**. Wait a few seconds - a typical answer takes 3 to 8 seconds.
4. Read the answer in the **Conversation** panel. Above it, badges show how many AI requests you have left today, which AI model answered and how many milliseconds it took.
5. Ask a **follow-up** such as 'explain that more simply'. The tutor remembers the conversation, so you do not need to repeat yourself.
6. Click **New chat** to wipe the conversation and start on a fresh topic. Your conversation is saved, so it is still there after you reload the page or sign in again.
7. Use the **Copy** button in the Raw JSON section if you want the raw response data.

Not sure what to ask? Click one of the **Try an Example** buttons and it fills the question box for you.

> **Tip:** Be specific. 'Explain derivatives in calculus with a real-world example' gives a much better answer than 'calculus'. The tutor remembers your recent messages (about the last 6 questions and answers), so click New chat when you switch subject - otherwise old context can influence the answer.

3.3 Study Notes

1. Click **Study Notes**.
2. Paste your lecture notes or textbook text into the box (up to 10,000 characters).
3. Choose what you want: **Summarize**, **Key Points**, **Explain Simply**, **Generate Quiz** (a 3-question quiz with answers) or **Flashcards** (5 front/back cards).
4. Wait for the result to appear, then use the copy button to copy it.

Each result is **saved automatically** to the **Saved Notes** list on the left. Click a saved note to reopen it, and use its delete button to remove it.

3.4 Quiz Generator

1. Click **Quiz Generator**.
2. Fill in at least one of: **Topic**, **Study Material** (pasted text) or **Upload PDF**. You can combine them.
3. Pick a **Difficulty** (Easy, Medium or Hard) and the **Number of Questions** (5, 10 or 15).
4. Click **Generate Quiz**.
5. For each question, choose one of the four options and click **Submit Answer**. You immediately see whether you were right, with an explanation.
6. Click **Next Question** until the end. You then see your score and percentage.

Your result is **saved automatically** when you finish and feeds the dashboard's quiz count, average score, streak and activity chart.

> **Important:** PDF upload reads the text inside the PDF. Scanned PDFs (photos of pages) contain no selectable text and will not work. Only files ending in .pdf are accepted.

3.5 Flashcards

1. Click **Flashcards**.
2. Enter a **Topic** and/or paste **Study Material**, and pick the **Number of Cards** (5, 10, 15 or 20).
3. Click **Generate Deck**.
4. Click a card to flip it between the question (front) and the answer (back). Use the left/right arrows to move between cards.
5. Use **Shuffle** to randomise the order, or **New Deck** to make another one.

Decks are **saved automatically** to **Saved Decks** on the left, so you can reopen or delete them later.

3.6 Study Planner

1. Click **Study Planner**.

2. In the **Add Task** form fill in **Subject**, **Task Description**, **Due Date** and **Priority** (High, Medium or Low; Medium is the default).
3. Click **Add Task**. It appears in your task list.
4. Tick a task when you finish it. Use the trash icon to delete one.

The list is sorted so unfinished tasks come first, then by earliest due date, then by priority (High before Medium before Low). Completing a task raises your dashboard 'Tasks completed' counter and counts toward your streak and weekly activity on the day you completed it. Tasks due today also appear on the dashboard, where you can tick them off too.

3.7 Settings

- **Appearance:** choose Light or Dark mode, and switch on **Compact Layout** to fit more on screen. The moon/sun button in the top bar also toggles the theme. These two choices are remembered in your browser only.
- **Notifications:** switch on **Study reminders** to get a reminder once a day about tasks that are due today or overdue, and (after 6 pm) a nudge to keep your streak alive if you have not studied yet. Your browser asks permission the first time; if notifications are blocked, the reminder appears inside the page instead. Use **Test** to preview one. Reminders appear while StudyAI is open in a browser tab, and the choices are saved on this device only.
- **Account:** see your role (Student or Admin), change your password (you need the current one) and create or replace your recovery code. A new code makes the old one stop working, and it is shown only once.
- **Data & Storage: Clear Data** permanently deletes all your tasks, quiz results, notes, flashcard decks and AI Tutor conversation. Your account stays. This cannot be undone.
- **Privacy:** a short statement on how your data is handled.

4. Limits and small details

Item	Detail
Username and password	Username 3 to 32 characters; password 6 to 128 characters. Usernames are unique.
AI Tutor question length	Up to 2,000 characters.
Notes / quiz / flashcard input	Up to 10,000 characters in total (topic + pasted material + PDF text).
Quiz size	5, 10 or 15 questions; each has 4 options, one correct answer and an explanation.
Flashcard deck size	5, 10, 15 or 20 cards.
Notes actions	Summarize, Key Points, Explain Simply, Generate Quiz (3 questions), Flashcards (5 cards).
Saved automatically	Notes results, flashcard decks and finished quiz results. Planner tasks save when you click Add Task.
Privacy between users	Every record belongs to one account; other users cannot open it.
Staying signed in	You stay signed in until you click Logout or close the browser.
Theme and layout choices	Stored in your browser, so they do not follow you to another device.
AI Tutor memory	Your recent messages are stored and sent with each new question (up to 12 messages, 60 stored). New chat or Clear Data erases them.
AI usage limits	Each user gets 60 AI requests per rolling 24 hours and 6 per minute, across the Tutor, Notes, Quiz and Flashcards. The Tutor shows how many are left. The owner can change these.
Sign-in history	Account creation, sign-ins and failed attempts are recorded with the network address and browser for 90 days. Only the app administrator can see them.
Sign-in protection	5 wrong passwords for one username pause sign-in for 15 minutes; many failures from one network address are also slowed. At most 20 new accounts per network address per hour.
Where AI runs	Your text is sent to Google's Gemini API to produce answers. Avoid pasting passwords or private personal data.

Tips for better results

- Give the AI context: level, subject and what you want ('explain like I am a first-year student').
- For quizzes, paste your own notes instead of only a topic - questions then match what you actually study.
- Keep pasted text under the limit; split very long chapters into sections.
- Check important facts against your textbook. AI answers can occasionally be wrong.

5. Troubleshooting

What you see	What it means / what to do
Rate limit exceeded. Please try again later.	The free AI quota is used up for now. Wait a minute and retry; if it persists all day, the app owner needs to add another API key (see Part 8).
Invalid API key / API Key is invalid or unauthorized	The owner's Gemini key is wrong or was deleted. Owner: update the key (see Part 8).
Gemini API is not configured	No key was found on the server. Owner: set GEMINI_API_KEY and restart or redeploy.
The response was blocked by safety settings	Gemini refused that content. Rephrase the question.
The request timed out	The AI took too long. Try again, or ask a shorter question.
AI generated invalid data format	The quiz/flashcard response was not readable. Click Generate again.
Failed to read the PDF file	The file is damaged, password-protected or scanned images. Use a text-based PDF, or paste the text instead.
Invalid username or password	Check spelling and caps lock. Usernames and passwords are case-sensitive.
You've used all 60 of your AI requests for today / limit is 6 AI requests per minute	This is your personal limit, not an outage. Wait for the time shown (it counts down from your oldest request) and try again.
Too many failed sign-in attempts	Wait the time shown (up to 15 minutes), or use Forgot password? with your recovery code, which lifts the pause.
That username and recovery code don't match	Check the username and the 16-character code. Each code works once: after a reset or a new code, only the newest one is valid.
Security check failed. Please refresh the page / Your session expired	Your session or page went stale (for example the tab was left open a long time, or cookies are blocked). Reload the page and try again.
Too many accounts were created from this network	A limit of 20 sign-ups per hour per network applies. Wait and try again.
Password must be at least 6 characters / Passwords do not match	Fix the sign-up form: use 6 or more characters and type the same password in both boxes.
Sent back to the Sign In page	Your session ended (browser closed or logout). Sign in again.
Internal Server Error (owner)	Look at the deployment logs. The usual causes are a missing DATABASE_URL/Postgres database or a missing environment variable (see Part 7).

6. Owner guide: run it on your computer

Requirements: Python 3.10 or newer and a Gemini API key from https://aistudio.google.com/app/apikey (free).

1. Open a terminal in the project folder.
2. Create a virtual environment: `python -m venv venv`
3. Activate it on Windows: `venv\Scripts\activate` (macOS/Linux: `source venv/bin/activate`).
4. Install everything: `pip install -r requirements.txt`
5. Open the file named `.env` and set `GEMINI_API_KEY=` to your key and `SECRET_KEY=` to a long random string.
6. Start the app: `python app.py`
7. Open `http://localhost:5000` in your browser. If port 5000 is busy, set `PORT=5001` (any free port) in `.env` and use that port instead.

A minimal `.env` looks like this:

```
SECRET_KEY=change-this-to-a-long-random-string
GEMINI_API_KEY=your-gemini-key-here
DATABASE_URL=study_assistant.db
```

Tip: Locally the app stores data in a small file called study_assistant.db (SQLite). The tables are created automatically the first time the app starts - there is nothing to set up.

Important: Never share your .env file or post your API key online, and never commit it to GitHub. The project's .gitignore already excludes .env and the .db file. If a key is ever exposed, delete it in Google AI Studio and create a new one.

7. Owner guide: deploy on Railway so others can use it

Once deployed, anyone with the web address can create their own account and use the app at the same time. Railway runs the app with gunicorn using the included `Procfile` (`web: gunicorn app:app`).

Why a database is needed

A SQLite file on Railway is wiped every time the app redeploys or restarts, which deletes all users and data. Production must use a PostgreSQL database instead. The app switches to PostgreSQL automatically when `DATABASE_URL` starts with `postgresql://`, and creates all tables on startup.

Steps

1. Push your code to GitHub and connect the repository to a Railway project (New Project, Deploy from GitHub repo).
2. In the same Railway project click **New**, then **Database**, then **Add PostgreSQL**.
3. Open your web service, then the **Variables** tab. Make sure `DATABASE_URL` is available to the web service. Railway can reference the database's URL for you (Add Reference, then pick the Postgres `DATABASE_URL`).

4. In **Variables** also add `GEMINI_API_KEY` and `SECRET_KEY` (your local .env file is *not* uploaded, so these must be set in Railway).
5. Under **Settings, Networking** click **Generate Domain** to get the public web address.
6. Redeploy. Open the web address, register an account, and try the AI Tutor.
7. To check health, open `/api/health` on your domain. It should show `"status": "ok"` and `"gemini_ready": true`.

Important: Keep SECRET_KEY the same forever. If it changes, every logged-in user is signed out.

Tip: Cloudflare Workers/Pages cannot run this Flask + Python app without a full rewrite. Use Railway (or a similar Python host such as Render). Cloudflare can still sit in front of it as a DNS/CDN layer.

8. Owner guide: roles, accounts and sign-in history

Every account has one of two roles. The role is stored in the database and checked on every request, so a change (promotion, demotion, suspension) takes effect immediately, even for people who are already signed in.

Role	What it can do
Student	Everything in the study app: AI Tutor, Notes, Quiz, Flashcards, Planner, Dashboard and Settings. A student only ever sees their own data, and the Admin page does not exist for them (it returns 'page not found').
Admin	Everything a student can do, plus the **Admin** page: see all accounts and sign-in history, change roles, suspend or reactivate accounts, unlock people, issue recovery codes and delete accounts.

Every new account starts as a Student. There is no way to become an admin by signing up.

Create the first admin

1. Create (or sign in to) your own account in the app first.
2. Locally, put `ADMIN_USERS=your-username` in `.env` (exact spelling, capital letters matter) and restart. On Railway, add the same variable under the web service's **Variables** and let it redeploy.
3. Sign in. An **Admin** link appears in the menu (address `/admin`), and Settings, Account shows 'Role: Admin'.

> **Tip:** `ADMIN_USERS` only works while there is **no admin yet**. Once one admin exists the setting is ignored, so nobody can become admin by registering a listed name later. After that, make admins on the Admin page.

> **Important:** Register your own account **before** setting ADMIN_USERS. Until the first admin exists, someone else could register the name you plan to use.

If you are ever locked out (for example the only admin is deleted from the database), run this in the project folder: `flask --app app make-admin YOUR-USERNAME` (on Railway, run it with the Railway CLI's `railway run`).

What an admin can do

Action	What it does
Make admin / Make student	Changes the role. The person's open session gets or loses access immediately.
Suspend / Reactivate	A suspended person is signed out at once and can't sign in. They are told the account is suspended only after entering the correct password. Their data is kept.
Unlock	Lifts a 'too many failed attempts' pause for that username.
Recovery code	Creates a new one-time code, shown once in a pop-up, for someone who lost both password and code. You hand it over and they set their own new password on the Forgot password page. Their old code stops working.
Delete	Permanently deletes the account and all its tasks, quizzes, notes, flashcards and tutor chat. Cannot be undone.

- You **cannot change your own account** (role, suspension, deletion). This guarantees at least one admin always remains; another admin must do it.
- Every admin action is written to the activity list with who did it, for example 'Role changed by admin - by alice: student -> admin'.
- Admins never see or choose anybody's password.

What the Admin page shows

- **Summary:** accounts (and how many admins), signed in in the last 24 hours, new in the last 7 days, failed sign-ins, suspended accounts.
- **Accounts table:** username, role, status, created date, last sign-in, number of sign-ins, tasks, quizzes, notes, decks, recovery-code status and the action buttons. Use the search box to find someone.
- **Recent activity:** the latest 150 events (account created, signed in, wrong password, blocked, password reset or changed, recovery codes, and every admin action) with time, network address and browser. Times are UTC.

Can the page show passwords?

No, and that is deliberate. Passwords are stored only as one-way hashes, so even an admin cannot read them. Showing passwords in plain text would be a **bad** option: anyone who gained access to the page, the database or a backup would get every user's password, and people reuse passwords on email and banking sites. The page shows a masked 'hashed' column instead.

Looking at the raw database

- **On your computer:** the data is in `study_assistant.db`. Open it with the free *DB Browser for SQLite* (tables `users`, `login_events`, `study_tasks` and so on). Close the app first, or open it read-only.
- **On Railway:** open the PostgreSQL service and use its **Data** tab.
- **MySQL is not needed.** The app uses SQLite locally and PostgreSQL online; switching to MySQL would mean rewriting the database code with no benefit.

Privacy and safety

- Sign-in history is kept for 90 days, then deleted automatically (`LOGIN_LOG_DAYS`). Settings, Privacy tells users that it is recorded.
- Addresses and browsers are personal data in many places. Keep admin accounts to people you trust and never share an admin login.

> **Important:** Failed attempts show the username that was typed, which is sometimes a mistyped password. Treat the activity list as sensitive.

9. Owner guide: Gemini API keys and free quota

The free Gemini tier has a limited number of requests per minute and per day. The quota belongs to your Google project, so it is **shared by every user** of your deployed app. Google does not allow an app to create new API keys by itself, so the app instead **switches automatically** between the keys and models you give it.

Change or replace your key

1. Go to https://aistudio.google.com/app/apikey and create a new key.
2. Locally: replace the value after `GEMINI_API_KEY=` in `.env` and restart the app.
3. On Railway: replace `GEMINI_API_KEY` in the web service's **Variables** tab. Railway redeploys.
4. Delete the old key in Google AI Studio once the new one works.

Add backup keys for automatic failover

Set `GEMINI_API_KEYS` to a comma-separated list. The app uses them in order and moves on to the next key when one runs out of quota or is invalid:

```
GEMINI_API_KEYS=new-key-1,new-key-2,new-key-3
```

- Create each key in a **different Google Cloud project**. Keys in the same project share one quota, so they do not help.
- `GEMINI_MODELS` (default `gemini-2.5-flash,gemini-3.5-flash-lite`) is tried in order too - each model has its own free quota, so a lighter model acts as a backup.
- A key/model that is out of quota is skipped for 60 seconds, then tried again. Invalid keys and unavailable models are skipped for 10 minutes.
- Users only see 'Rate limit exceeded' when *every* key and model is exhausted.
- Visit `/api/health` - `keys_configured` shows how many keys were loaded.

Checking how much quota is left

Gemini does not report a remaining-requests number through its API. View your usage and limits in the Google AI Studio usage page (https://aistudio.google.com/usage) or in the Quotas page of Google Cloud Console for the key's project.

> **Tip:** For heavy use (a whole class, for example), attach billing to your Google project for higher limits, or ask each user to use their own key. Per-user keys are not built into the app yet.

10. Project reference

Environment variables

Variable	Required	Purpose
SECRET_KEY	Yes	Signs login cookies. Use a long random value and never change it in production.
GEMINI_API_KEY	Yes*	Your Gemini API key. (*or use GEMINI_API_KEYS)
GEMINI_API_KEYS	No	Comma-separated backup keys used automatically when one runs out.
GEMINI_MODELS	No	Comma-separated model list tried in order.
DATABASE_URL	Yes (prod)	A postgresql:// address in production; a filename such as study_assistant.db locally.
PORT	No	Local port for python app.py (default 5000). Railway sets its own port for gunicorn.
SESSION_COOKIE_SECURE	No	Set to 1 to send login cookies over HTTPS only. Turned on automatically on Railway.
AI_LIMIT_PER_DAY / AI_LIMIT_PER_MINUTE	No	Per-user AI request caps (defaults 60 per day, 6 per minute; 0 turns a cap off).
LOGIN_MAX_FAILS / LOGIN_MAX_FAILS_IP / LOGIN_WINDOW_SECONDS	No	Sign-in lockout: failures per username (5) and per network address (30) within the window (900 s).
REGISTER_MAX_PER_HOUR	No	New accounts allowed per network address per hour (default 20).
ADMIN_USERS	No	First-admin setup only: exact usernames (comma-separated) promoted to admin at startup, and only while no admin exists yet.
LOGIN_LOG_DAYS	No	How many days of sign-in history to keep (default 90).
CHAT_CONTEXT_MESSAGES / CHAT_CONTEXT_CHARS / CHAT_MAX_STORED	No	How much Tutor conversation is sent to Gemini (12 messages, 12,000 characters) and kept per user (60).
TRUST_PROXY	No	Set to 1 when running behind another proxy so limits see real visitor addresses. Automatic on Railway.

Files

File / folder	Purpose
app.py	The whole backend: pages, login and security (CSRF, rate limits, recovery codes), database layer, Gemini calls and key failover.
templates/	HTML pages (home, login, register, forgot password, recovery code, dashboard, tutor, notes, quiz, flashcards, planner, settings, about).
static/	Stylesheet (css/style.css), browser scripts (js/main.js, js/notify.js for study reminders) and the favicon.
requirements.txt	Python packages: Flask, python-dotenv, google-genai, pypdf, gunicorn, psycopg.
Procfile	Production start command for Railway: web: gunicorn app:app
.env	Your private settings for local use. Never commit or share.
study_assistant.db	Local SQLite database file (created automatically; ignored by git).

Known limitations

- Passwords are reset with a recovery code, not by email (the app has no email service). A user with no saved code who forgets their password must make a new account.
- Sign-in pauses are per username, so someone could deliberately trigger a 15-minute pause on another person's username; a password reset with the recovery code lifts it.
- The free Gemini quota is still shared by everyone. Per-user caps and the sign-up limit stop one person using it all, but a large class still needs extra keys or billing.
- Reminders work only while StudyAI is open in a browser tab (there are no background push notifications), and their settings are saved per device.
- Extra protections (CSRF tokens, secure cookies) are built in; the app has no per-user Gemini key yet.
- There are only two roles (Student and Admin) and every admin has the same powers. Sign-in history starts from the moment that feature was added.

End of guide.